COMPLETE AWS OFFICIAL CONSOLE WALKTHROUGH

Amazon RDS Step-by-Step Database Setup Manual

Comprehensive guide covering all 9 configuration stages from console search to production connectivity with real AWS UI screenshots.

1 Navigate to the Amazon RDS Service in the AWS Console

Sign in to your AWS Management Console. Ensure your designated AWS Region (such as `us-east-1`) is selected in the top navigation bar. In the unified search box, enter RDS, click **Amazon RDS** under Services, and click the orange **Create database** button.

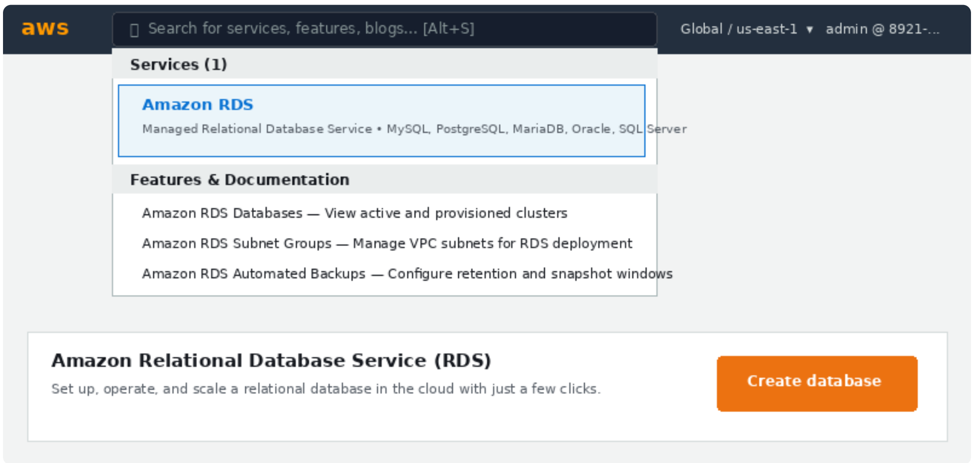

FIGURE 1: AWS CONSOLE UNIFIED SEARCH BAR AND RDS SERVICE LAUNCH DASHBOARD

- **Region Check:** Always verify your active AWS Region in the header before spinning up databases.
- **Direct Path:** You can also access database provisioning by clicking **Databases** in the left-hand navigation pane and selecting **Create database**.

2 Choose Database Creation Method (Standard Create)

On the **Create database** page, AWS presents two creation paths: **Standard create** and **Easy create**.

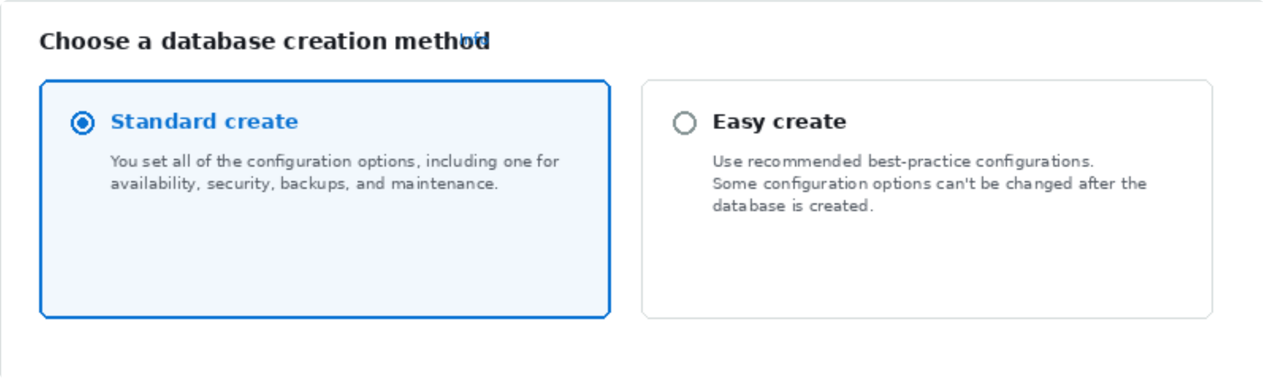

FIGURE 2: REAL AWS CONSOLE UI — DATABASE CREATION METHOD SELECTION

- **Standard create (Recommended):** Provides complete granular control over VPC subnets, public accessibility, automated backups, encryption KMS keys, and maintenance windows.
- **Easy create:** Uses pre-selected AWS defaults with minimal configuration, but restricts adjusting network topology or storage specifications later.

3 Select Database Engine & DB Instance Sizing Template

Under **Engine type**, select your target relational database engine. Then scroll to the **DB instance size** / **Template** card to specify your workload tier.

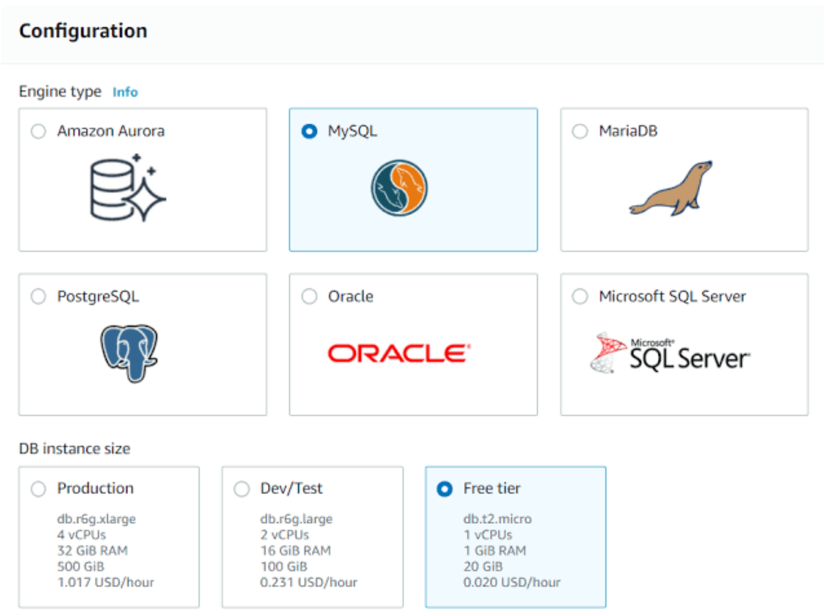

FIGURE 3: REAL AWS CONSOLE UI — RELATIONAL ENGINE TYPES & DB INSTANCE SIZE TIERS

- **Engine Options:** Choose **MySQL** (Community Edition), **PostgreSQL**, **Amazon Aurora**, **MariaDB**, **Oracle**, or **Microsoft SQL Server**.
- **Free tier:** Select **Free tier** to use a `db.t2.micro` or `db.t3.micro` instance (1 vCPU, 1 GiB RAM, 20 GiB storage) eligible under the AWS Free Usage Tier.
- **Production** / **Dev-Test:** Automatically provisions multi-core compute classes with Multi-AZ redundancy for high availability.

4 Configure Instance Identifier & Master Authentication

Define your database instance name and configure root administrative authentication credentials.

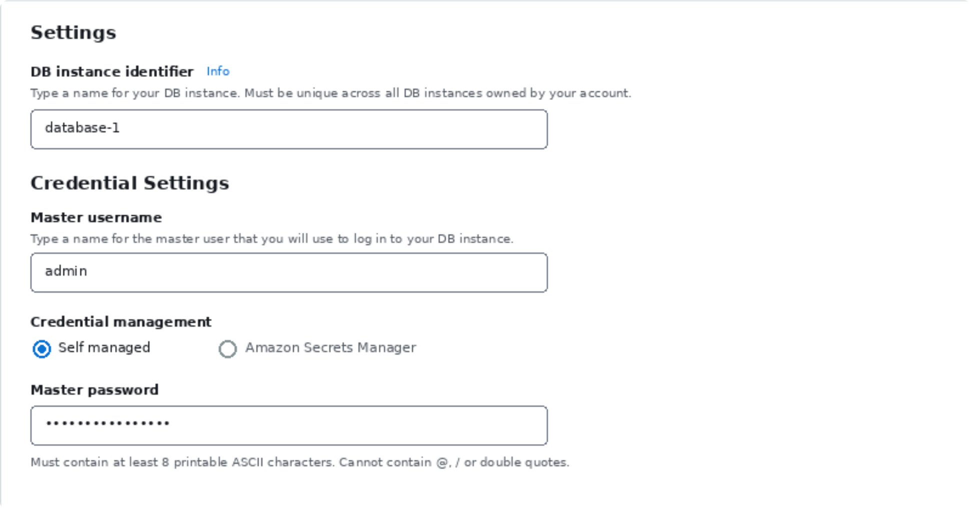

FIGURE 4: REAL AWS CONSOLE UI — DB INSTANCE IDENTIFIER AND CREDENTIAL SETTINGS

- **DB instance identifier:** Assign a unique alphanumeric name across your AWS account (e.g., `database-1`).
- **Master username:** The default administrator login (e.g., `admin`).
- **Credential management:** Choose **Self managed** to set a permanent password, or delegate to **Amazon Secrets Manager** for automated password encryption and rotation.

5 Configure DB Instance Class & Storage Sizing

Choose the compute instance class and configure the EBS storage volume type and automatic storage autoscaling.

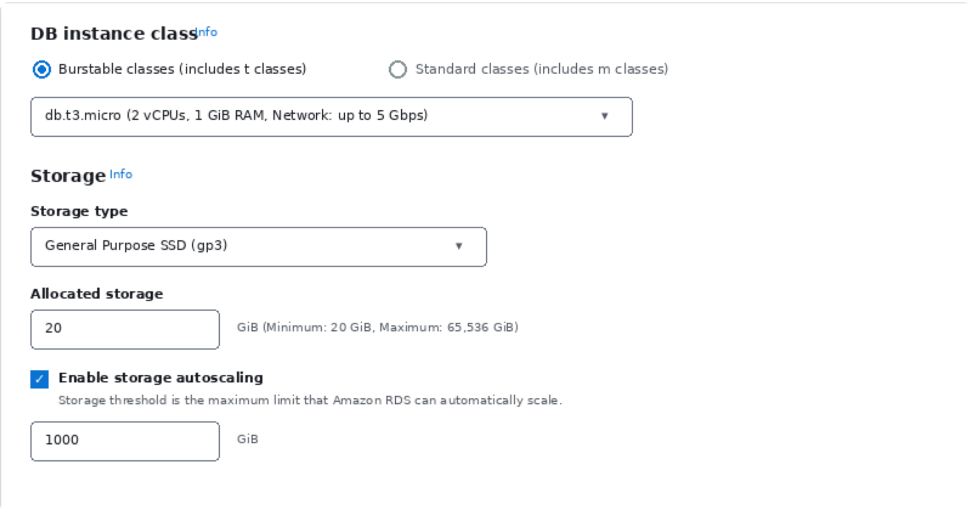

FIGURE 5: REAL AWS CONSOLE UI — COMPUTE INSTANCE CLASS AND STORAGE AUTOSCALING OPTIONS

- **DB instance class:** Select **Burstable classes** (e.g., `db.t3.micro` with 2 vCPUs and 1 GiB RAM) for development, or **Standard classes** (m5/m6g) for enterprise loads.
- **Storage type:** Select **General Purpose SSD (gp3)** for cost-effective baseline IOPS.
- **Allocated storage:** Set baseline storage (e.g., `20 GiB`) and enable **Storage autoscaling** with a ceiling (e.g., `1000 GiB`) to prevent disk-full outages.

6 Configure VPC Connectivity & Firewall Security Groups

Isolate database networking and define inbound firewall rules to govern which clients can establish connections.

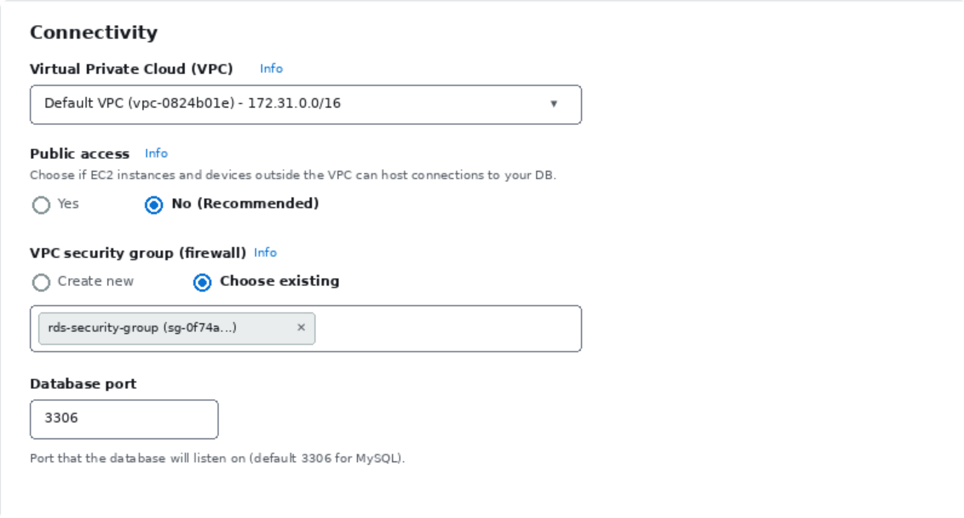

FIGURE 6: REAL AWS CONSOLE UI — VPC SELECTION, PUBLIC ACCESS TOGGLE, AND FIREWALL RULES

- **Virtual Private Cloud (VPC):** Select your application VPC (e.g., `Default VPC`).
- **Public access:** Select **No (Recommended)** to keep the database internal to your private VPC subnets.
- **VPC security group:** Attach a security group permitting inbound TCP traffic on port `3306` (MySQL) or `5432` (PostgreSQL) originating from your backend app servers.

7 Configure Additional Options (Backups, Encryption & Deletion Protection)

Expand the **Additional configuration** collapsible accordion to specify initial database creation, automated backup schedules, encryption, and accidental deletion safeguards.

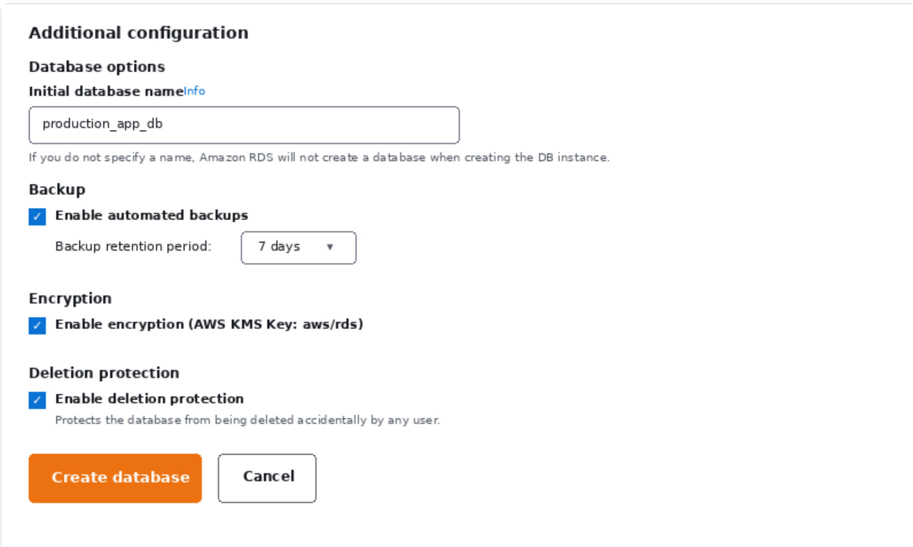

FIGURE 7: REAL AWS CONSOLE UI — INITIAL DATABASE NAME, BACKUP RETENTION, KMS ENCRYPTION & DELETION PROTECTION

- **Initial database name:** Specify the schema name (e.g., `production_app_db`) to be automatically initialized.
- **Backup:** Keep **Enable automated backups** checked with a `7 days` retention period.
- **Encryption:** Ensure **Enable encryption** is active using the default AWS KMS RDS key (`aws/rds`).
- **Deletion protection:** Check **Enable deletion protection** to prevent accidental database deletion.
- **Create database:** Click the orange **Create database** button at the bottom of the page.

8 Monitor Provisioning Status in the RDS Console

After initiating creation, you are returned to the RDS Databases list view. The instance will undergo initialization and health checks.

Databases (1)

DB identifier	Role	Engine	Region & AZ	Size	Status
database-1	Instance	MySQL Community 8.0.35	us-east-1a	db.t3.micro	Creating
database-1 (Ready)	Instance	MySQL Community 8.0.35	us-east-1a	db.t3.micro	Available

Provisioning takes ~5-10 minutes. Status shifts from 'Creating' to 'Available' once ready for connections.

FIGURE 8: REAL AWS CONSOLE UI — DATABASE STATUS PROGRESSION FROM 'CREATING' TO 'AVAILABLE'

- **Status Lifecycle:** The database status will display as **Creating** for approximately 5 to 10 minutes while compute and EBS storage are mounted.
- **Ready State:** Once health verification passes, the status changes to **Available**.

9 Retrieve Connection Endpoint, Port & Connect via Client

Click on your database identifier (`database-1`) to open its details panel. Navigate to the **Connectivity & security** tab to locate the DNS Endpoint and Port.

database-1 Available

MySQL Community 8.0.35 • us-east-1a • db.t2.micro • Free tier

Connectivity & security | Monitoring | Logs & events | Configuration | Maintenance & backups

Endpoint & port

Endpoint	Port
database-1.c418931b2.us-east-1.rds.amazonaws.com	3306

VPC: vpc-0824b01e | Subnet group: default-vpc-0824b01e | Publicly accessible: No

Security group rules (Active inbound)

Type	Protocol	Port Range	Source	Description
MySQL/Aurora	TCP	3306	sg-0e998124b (App-SG)	EC2 App Server inbound

FIGURE 9: REAL AWS CONSOLE UI — ACTIVE INSTANCE DNS ENDPOINT, PORT, AND SECURITY GROUP VERIFICATION

> **Connection Verification:** Copy the full **Endpoint** address. Connect from your application backend or local terminal using the credentials established in Step 4.

```
# 1. Connect via MySQL Command-Line Client:
mysql -h database-1.c418931b2.us-east-1.rds.amazonaws.com -P 3306 -u admin -p

# 2. Connect via PostgreSQL psql Client:
psql -h database-1.c418931b2.us-east-1.rds.amazonaws.com -p 5432 -U admin -d production_app_db
```